

Meysam Varasteh

Ph.D. Student in Computer Science | Recommender Systems & Interpretable Machine Learning

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Boulder, Colorado, USA

Research Interests

Recommender systems; interpretable and controllable machine learning; mechanistic interpretability, personalization and ranking

Education

University of Colorado Boulder

Aug. 2023 – Expected
May 2028

Ph.D. in Computer Science, Boulder, Colorado

- Advisor: Prof. Robin Burke.
- Thesis: Explainable AI in multi-stakeholder recommender systems

University of Tehran

2018 – 2021

M.Sc. in Electrical and Computer Engineering, Tehran, Iran

K.N.Toosi University

2014 – 2017

M.Sc. in Physics, Tehran, Iran

Petroleum University of Technology

2010 – 2014

B.Sc. Electrical and Computer Engineering, Iran

Research Experience

Graduate Research Assistant

2023 – Present

University of Colorado Boulder

Research Assistant

2021-2022

Polytechnic University of Bari

Graduate Research Assistant

2018-2021

University of Tehran

Graduate Research Assistant

2015-2018

K.N.Toosi University

Publications and Manuscripts

Manuscript in preparation

1. **Meysam Varasteh** et al. “From Tokens to Concepts: A Post-hoc Concept Bottleneck for Interpretable and Steerable Recommendation.” Manuscript in preparation.

Peer-reviewed papers

2. **Meysam Varasteh**, Robin Burke. “Why Didn’t More People See It? Recommendation Transparency for Providers.” *IntRS Workshop @ ACM RecSys 2026*. [\[paper\]](#)
3. **Meysam Varasteh**, Veronika Bogina, Noam Koenigstein, Robin Burke. “Why This, Not That? Mining User Profiles for Pair-wise Counterfactuals.” *IntRS Workshop @ ACM RecSys 2026*. [\[paper\]](#)
4. **Meysam Varasteh**, Elizabeth McKinnie, Amanda Aird, Daniel Acuña, Robin Burke. “Comparative Explanations for Recommendation: Research Directions.” *11th Joint Workshop on Interfaces and Human Decision Making for Recommender Systems (IntRS @ RecSys 2024)*, 2024, pp. 1–14. [\[paper\]](#)
5. **Meysam Varasteh**, Mehdi Soleiman Nejad, Hadi Moradi, Mohammad Amin Sadeghi, Ahmad Kalhor. “An Improved Hybrid Recommender System: Integrating Document Context-Based and Behavior-Based Methods.” *31st International Conference on Electrical Engineering (ICEE)*, 2023. [\[IEEE\]](#)

Preprints and research manuscripts

6. Mehdi Soleiman Nejad, **Meysam Varasteh**, Hadi Moradi, Mohammad Amin Sadeghi. “Designing a Sequential Recommendation System for Heterogeneous Interactions Using Transformers.” *arXiv preprint arXiv:2205.00265*, 2022. [arXiv]

7. Yashar Deldjoo, **Meysam Varasteh**, Yara M. Bahram, Marco Antonio Insabato, Tommaso Di Noia. “Fairness Auditing Through the Lens of Counterfactual Explanations.” Research manuscript / project on counterfactual fairness auditing.

Teaching Experience

Teaching Assistant – CSCI 5434: Probability for Computer Science <i>University of Colorado Boulder</i>	Spring 2024 - 2026
Teaching Assistant – CSCI 2400: Computer Systems <i>University of Colorado Boulder</i>	Fall 2023-2026
Teaching Assistant – Neural Networks and Deep Learning <i>University of Tehran</i>	Fall 2018-2020
Teaching Assistant – Statistical Inference <i>University of Tehran</i>	Spring 2019-2020

Mentorship and Service

Mentor for Undergraduate Mentees – CSCI 4830 <i>University of Colorado Boulder</i>	2026
Student Volunteer <i>ACM SIGIR</i>	2024
Student Volunteer <i>ACM RecSys</i>	2023
Data Mining Workshop Organizer <i>University of Tehran</i>	2022

Honors and Awards

2026	Outstanding Teaching Assistant Award, Department of Computer Science, University of Colorado Boulder.
2026	WSDM NSF Conference Travel Grant, National Science Foundation.
2025	Departmental Summer Research Fellowship, Department of Computer Science, University of Colorado Boulder.
2024	SIGIR Conference Student Travel Grant, ACM SIGIR.
2023	Dean’s and Departmental Excellence Fellowship, University of Colorado Boulder.
2023	Early Career Professional Development Fellowship, University of Colorado Boulder.

Industry Experience

Data Scientist <i>Maktabkhooneh – Online Learning Platform</i>	Aug. 2018 – Feb. 2020
Designed a sequential recommender system to personalize educational content recommendations from learners’ historical interactions.	

Technical Skills

Programming	Python, C++, JavaScript
ML/Data	PyTorch, TensorFlow, Hugging Face, scikit-learn, CUDA
Tools	Git, Docker, Podman, Linux, LaTeX
Research	RecBole, experimental evaluation, reproducible ML pipelines